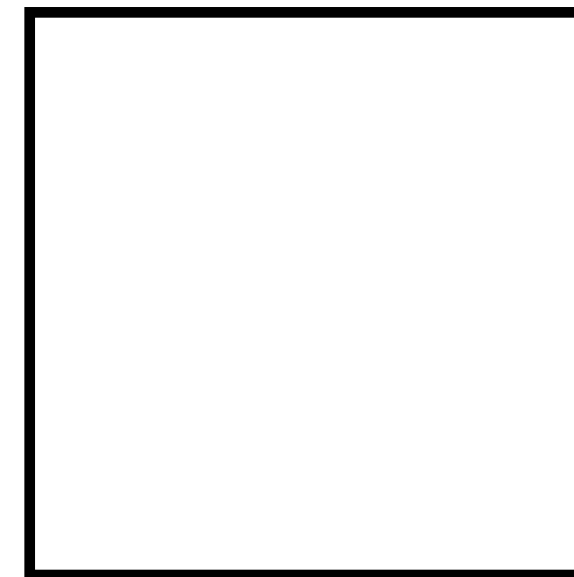


# An uncertainty principle for entanglement



Mitali Nanda



Daochen Wang

# Uncertainty principle

$$\Delta x \Delta p \geq \hbar/2$$

**In general: for  $\mathcal{A} = \{e_1, \dots, e_n\}$  and  $\mathcal{B} = \{f_1, \dots, f_n\}$  related by Fourier transform**

$$H(\mathcal{A}) + H(\mathcal{B}) \geq c$$

H quantifies outcome uncertainty

# Uncertainty induced by entanglement in $\mathcal{A}, \mathcal{B}$

$\mathcal{A}$  = Bell basis

$\mathcal{B}$  = computational basis

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$|00\rangle$$

$$\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$

$$|01\rangle$$

$$\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

$$|10\rangle$$

$$\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

$$|11\rangle$$

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

$$|00\rangle = \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)\right) + \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)\right)$$

states maximally entangled

states unentangled



# Quantifying uncertainty

**Measuring**  $b_{00}|00\rangle + b_{01}|01\rangle + b_{10}|10\rangle + b_{11}|11\rangle$  **in basis**  $\mathcal{B}$ :

A measure of uncertainty: Shannon entropy  $H(\mathcal{B}) = - \sum_x |b_x|^2 \log_2(|b_x|^2)$

## Examples

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

$$H(\mathcal{B}) = - (1/2)\log_2(1/2) - (1/2)\log_2(1/2) = 1$$

$$H(\mathcal{A}) = 0$$

$$|00\rangle = \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)\right) + \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)\right)$$

$$H(\mathcal{B}) = 0$$

$$H(\mathcal{A}) = - (1/2)\log_2(1/2) - (1/2)\log_2(1/2) = 1$$

# Maassen-Uffink entropic uncertainty relation

For *any* state and orthonormal bases  $\mathcal{A} = \{e_1, \dots, e_n\}$  and  $\mathcal{B} = \{f_1, \dots, f_n\}$ :

$$H(\mathcal{A}) + H(\mathcal{B}) \geq -2 \log_2(\max_{i,j} |\langle e_i | f_j \rangle|)$$

**Example**  $\mathcal{A} = \frac{1}{\sqrt{2}}\{ |00\rangle + |11\rangle, |00\rangle - |11\rangle, |01\rangle + |10\rangle, |01\rangle - |10\rangle \}$

$$\mathcal{B} = \{ |00\rangle, |01\rangle, |10\rangle, |11\rangle \}$$

$$\max_{i,j} |\langle e_i | f_j \rangle| = 1/\sqrt{2} \implies H(\mathcal{A}) + H(\mathcal{B}) \geq 1$$

# Quantifying entanglement

Take  $\psi \in \mathbb{C}^d \otimes \mathbb{C}^d$

**Fact:** can always write  $\psi = \sum_i \lambda_i u_i \otimes v_i$  such that

$\lambda_i > 0$ ,  $\{u_i\}_i$  orthonormal, and  $\{v_j\}_j$  orthonormal (Schmidt decomposition)

**Entanglement entropy of  $\psi$ :**  $E(\psi) = -\sum_i \lambda_i^2 \log_2 \lambda_i^2$

**Fact:**  $\lim_{\alpha \rightarrow 1} E_\alpha = E$

**Entanglement Rényi entropy of  $\psi$ :**  $E_\alpha(\psi) = \frac{1}{1-\alpha} \log_2 \left( \sum_i \lambda_i^{2\alpha} \right)$ ,  $\alpha > 0; \alpha \neq 1$



# Uncertainty relation induced by entanglement

For bases  $\mathcal{A} = \{e_1, \dots, e_n\}$  and  $\mathcal{B} = \{f_1, \dots, f_n\}$  of bipartite space and state  $\psi$

Write  $\psi = \sum_i a_i e_i = \sum_j b_j f_j$ , then for all  $\alpha \in (1, \infty]$  and  $\beta \in [1/2, 1)$ ,

$$\frac{\alpha}{\alpha - 1} H(\mathcal{A}) + \frac{\beta}{1 - \beta} H(\mathcal{B}) \geq \sum_i |a_i|^2 E_\alpha(e_i) - \sum_j |b_j|^2 E_\beta(f_j)$$

Uncertainty in measuring  
in bases  $\mathcal{A}$  and  $\mathcal{B}$

Difference in the entanglement Rényi  
entropies of basis states in  $\mathcal{A}$  and  $\mathcal{B}$

# Recovering special case

For bases  $\mathcal{A} = \{e_1, \dots, e_n\}$  and  $\mathcal{B} = \{f_1, \dots, f_n\}$  of bipartite space and state  $\psi$

Write  $\psi = \sum_i a_i e_i = \sum_j b_j f_j$ , then for all  $\alpha \in (1, \infty]$  and  $\beta \in [1/2, 1)$ ,

$$\frac{\alpha}{\alpha - 1} H(\mathcal{A}) + \frac{\beta}{1 - \beta} H(\mathcal{B}) \geq \sum_i |a_i|^2 E_\alpha(e_i) - \sum_j |b_j|^2 E_\beta(f_j)$$

**Example**

$$\mathcal{A} = \frac{1}{\sqrt{2}} \{ |00\rangle + |11\rangle, |00\rangle - |11\rangle, |01\rangle + |10\rangle, |01\rangle - |10\rangle \}$$

$$\mathcal{B} = \{ |00\rangle, |01\rangle, |10\rangle, |11\rangle \}$$

$$E_\alpha(e_i) = \frac{1}{1 - \alpha} \log_2(1/2^\alpha + 1/2^\alpha) = 1 \text{ and } E_\beta(f_j) = 0$$

Take  $\alpha = \infty, \beta = 1/2$ :  
 $H(\mathcal{A}) + H(\mathcal{B}) \geq 1$



# Proof via entanglement of superpositions

**Question:** how does the entanglement of a superposition of states relate to the entanglement of the states being superposed?

First considered by Linden, Popescu, Smolin (PRL 06), and later studied by Gour (PRA 07) and Gour and Roy (PRA 08), and others

## Examples

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

can create entanglement by superposing unentangled states

$$|00\rangle = \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)\right) + \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)\right)$$

can destroy entanglement by superposing entangled states

# New bounds on entanglement of superpositions

For bases  $\mathcal{A} = \{e_1, \dots, e_n\}$  and  $\mathcal{B} = \{f_1, \dots, f_n\}$  of bipartite space and state  $\psi$

Write  $\psi = \sum_i a_i e_i = \sum_j b_j f_j$ , then for all  $\alpha \in (1, \infty]$  and  $\beta \in [1/2, 1)$ ,

$$E(\psi) \geq \sum_i |a_i|^2 E_\alpha(e_i) - \frac{\alpha}{\alpha - 1} H(\mathcal{A})$$

$$E(\psi) \leq \sum_j |b_j|^2 E_\beta(f_j) + \frac{\beta}{1 - \beta} H(\mathcal{B})$$

# Physics application

Let  $\psi$  be an MPS state with bond dimension  $D$  or a linear combination of  $D$  product states, then  $E(\psi) \leq \log_2 D$

Let  $\mathcal{A} = \{e_1, \dots, e_n\}$  be the energy eigenstates of a Hamiltonian

Then  $E(\psi) \geq \sum_i |a_i|^2 E_\alpha(e_i) - \frac{\alpha}{\alpha-1} H(\mathcal{A})$  implies

$$H(\mathcal{A}) \geq (1 - 1/\alpha) \left( \sum_i |a_i|^2 E_\alpha(e_i) - \log_2 D \right)$$

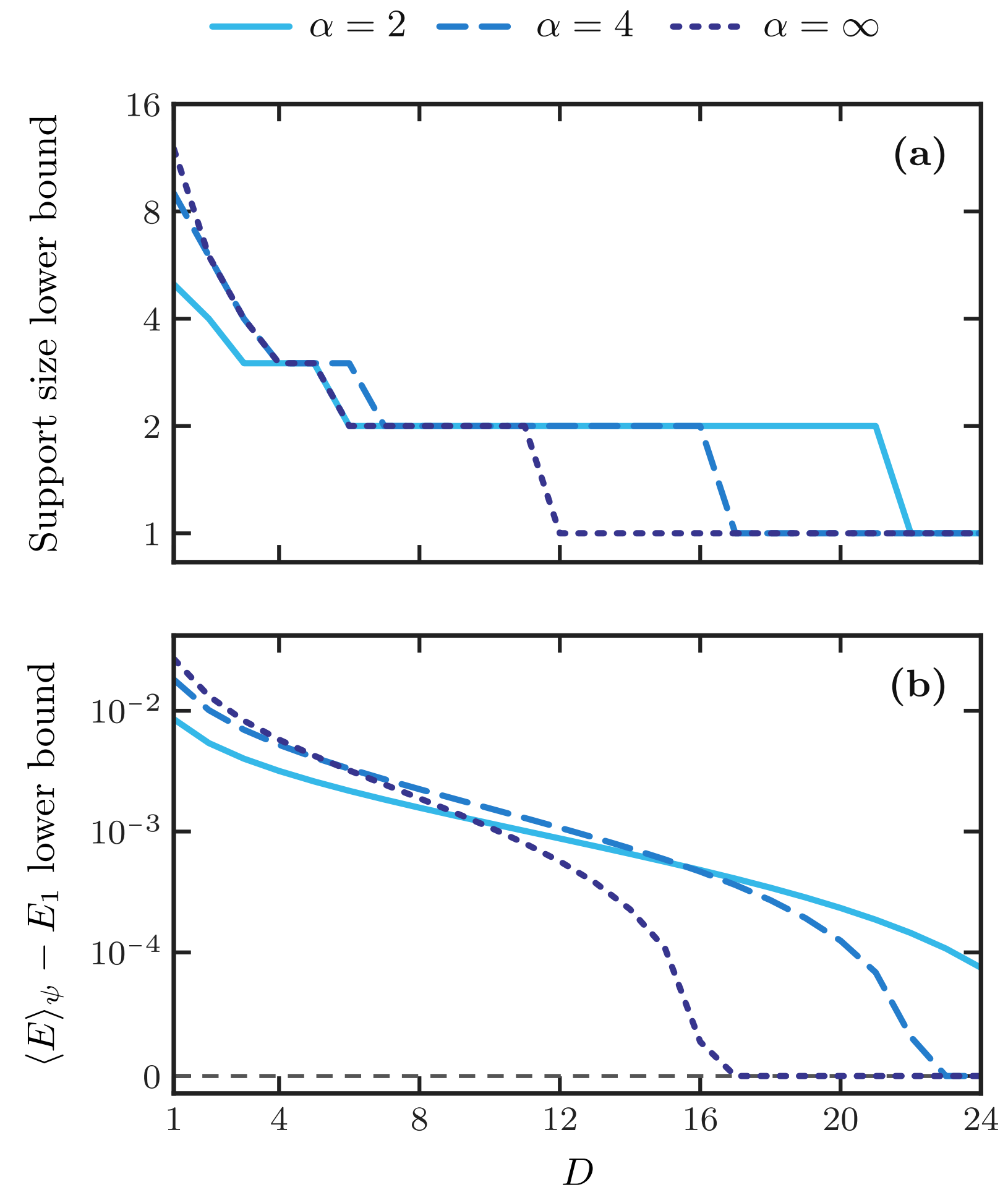
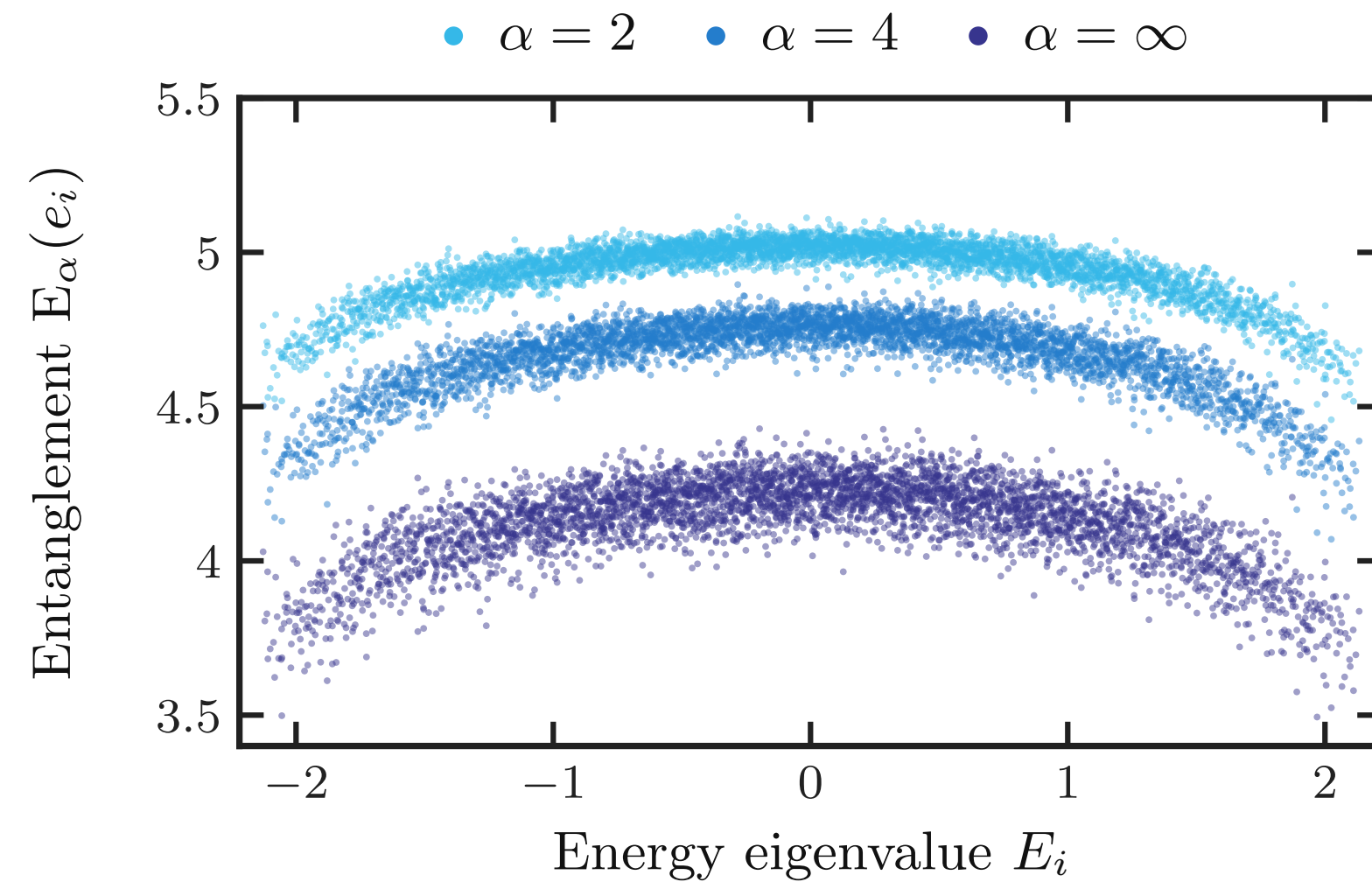
$\psi$  cannot be too certain in the energy eigenstate basis



# SYK model numerics

$$H_r^{\text{SYK}} = i^{r/2} \binom{n}{r}^{-1/2} \sum_{j_1 < \dots < j_r} g_{j_1 \dots j_r} \gamma_{j_1} \dots \gamma_{j_r},$$

$$(r = 4, n = 24)$$



# Open questions

1. Uncertainty principle for magic (or other resources)
2. Generalization to mixed states
3. Characterize max over  $\alpha$ , min over  $\beta$  versions of bounds
4. Generalization to multiple partitions

Thank you for  
your attention!